
Lecture 8

ECE 4107 (Section A)

Telecommunication Engineering



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Essential Features of Common Control

- Exchange must have capable of receive/store the digit dial
- Translate the exchange identifier into routine digits
- Transmit the routine and subscriber line identifier digits to switch network



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Common Control Switching System

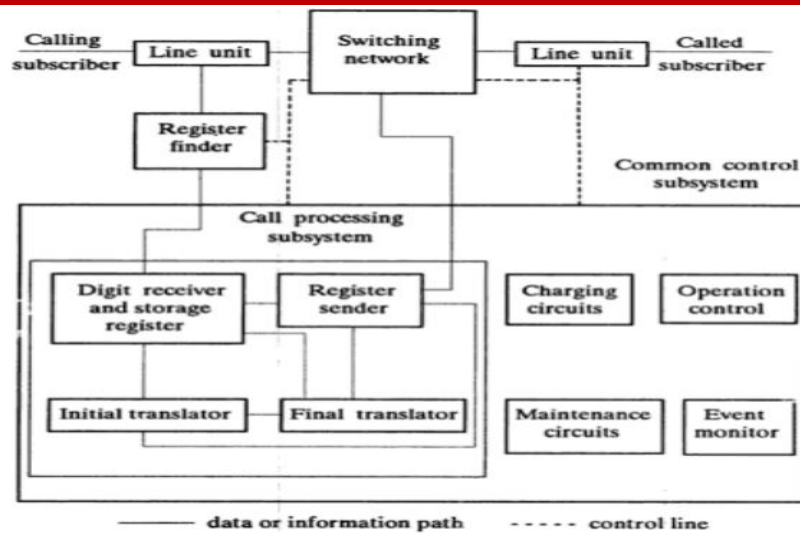


Fig. 3.2 Common control switching system.



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□ Diagram of Common Control Subsystem

■ Call processing subsystem

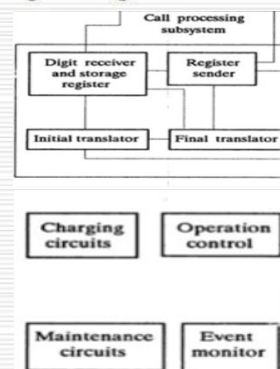
- Digit receiver and storage register
- Register Sender
- Initial translator
- Final translator

■ Charging circuits

■ Operation Control

■ Maintenance circuits

■ Event monitor



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Common control

□ Control Functions

■ Event Monitoring

- Events at line units
- Events at Trunk junctors
- Events at interexchange signaling receiver/sender units



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Common control

□ Control Functions

■ Call Processing

- Digit receiver and storage register: receive and store the dialing number from calling party.
- Initial translator (**office code translator**): determine the route for the call through the network, charging method/rate.
- Final translator (**subscriber code translator**): determine the line unit to which a call must be connected and category of the called line.
- Register sender: transfer the route digit and dialed digit using proper signaling.



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Common control

□ Control Functions

■ Charging

- Depends on the type of subscriber
- Depends on the service of the subscriber

■ Operation and Maintenance

- Controlling of the switching network
'Map-in-memory' & 'Map-in-Network'
- Administration & Maintenance
exchange equipment, subscriber lines, trunks



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3.2 Touch Tone Dial Telephone

□ Disadvantages of Rotary dial Telephone

- Low dialing rate
- Limited usage
- Limited signaling capacity

□ Development of Touch tone dial phone

- Developed in 1950s
- Applied in 1964.

□ Touch dial arrangement

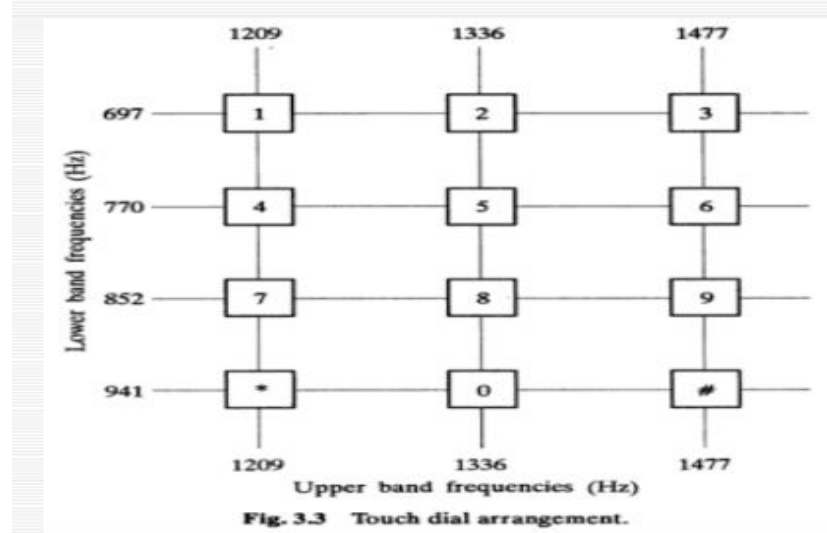
- DTMF: Dual Tone Multi-Frequency



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Touch Dial Arrangement



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Touch Tone Design Method

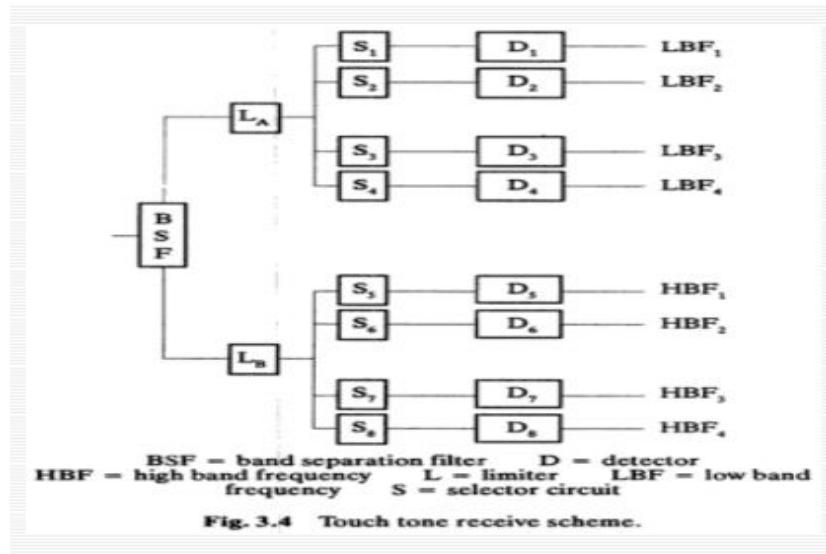
- Choice of code
- Band separation
- Choice of frequency
- Choice of power level
- Signaling duration



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Touch Tone Receiver



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Design Considerations

- ❑ Choice of code
 - Imitation of code signal by speech and music should be difficult.
- ❑ Band Separation
 - Easy to recognize a specific frequency
 - Easy to regulate
 - Reduce the probability of false response



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Design Considerations

- ❑ Choice of frequencies
 - Attenuation and delay distortion characteristics of the telephone network circuits
- ❑ Choice of power level
 - High enough for reliability
 - Power level should be planned according to attenuation characteristics of the channel
- ❑ Signaling duration
 - Longer duration is helpful to combat talk-off, but inefficient.
 - Dialing habit



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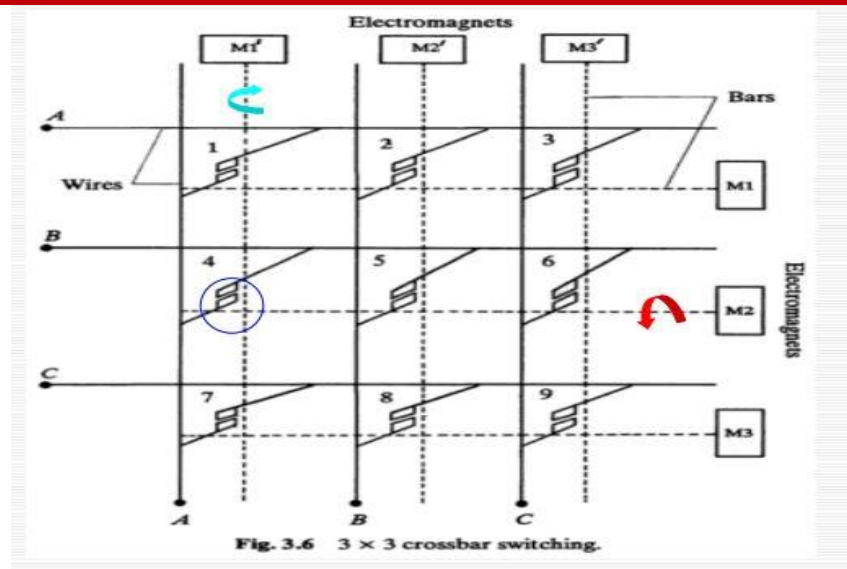
3.3 Principles of Crossbar Switching

- ❑ Basic idea
 - To provide a matrix of $(n \times m)$ sets of contacts with only $(n + m)$ activators or less to select one of the $(n \times m)$ sets of contacts.
- ❑ Crosspoint Switching matrix
 - Horizontal & Vertical wires
 - Contact points
 - Horizontal & Vertical bars
 - Electromagnets



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Crosspoint Switching matrix

□ Establishment of a connection

■ Scheme1

- Step1: Energize horizontal bar
- Step2: Energize vertical bar
- De-energize horizontal bar

■ Scheme2

- Step1: Energize vertical bar
- Step2: Energize horizontal bar
- De-energize vertical bar

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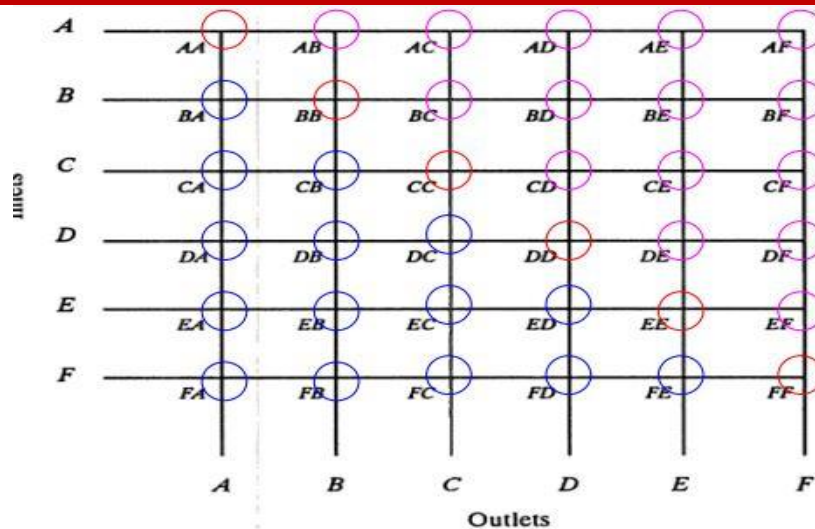


Fig. 3.7 6 × 6 crossbar matrix.



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3.4 Crossbar Switch Configurations

□ Nonblocking Crossbar configuration

- N^2 switching elements for N subscribers.
- $N/2$ simultaneous conversations.
- Crosspoint used depends on the calling ~.

Table 3.1 Nonblocking Crosspoint Switch Systems: Design Parameters

N	S	SC	EU	C	CCI
4	16	2	12.50	16	0.5
16	256	8	3.13	256	0.5
64	4096	32	0.78	4096	0.5
128	16384	64	0.39	16384	0.5

N = No. of subscribers S = No. of switching elements
 SC = switching capacity C = total cost
 EU = equipment utilisation factor CCI = cost capacity index



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3.4 Crossbar Switch Configurations

□ Modified Nonblocking Scheme

$$N^2/2$$

- Diagonal Crosspoint Matrix
- $N(N-1)/2$ elements
- The number of elements is the same as that of full-connected network.
- Connection establishment
 - Energize the corresponding horizontal bar
 - Energize the corresponding vertical bar



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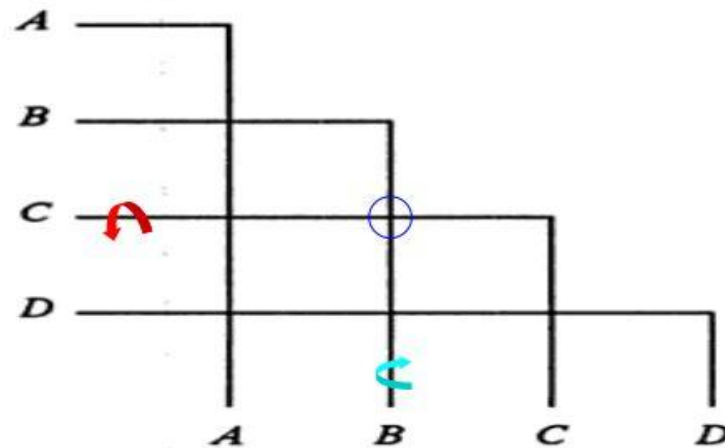


Fig. 3.8 Diagonal crosspoint matrix .



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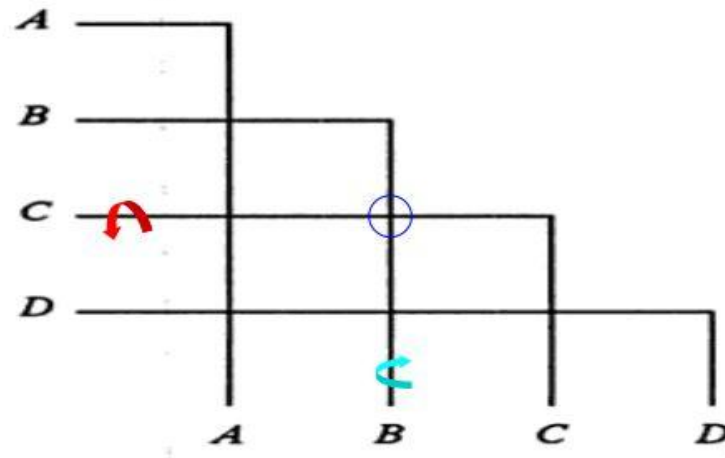


Fig. 3.8 Diagonal crosspoint matrix .



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3.4 Crossbar Switch Configurations

- Comments on nonblocking schemes
 - Large number of switching elements
 - Difficult to implement in practice
 - Not cost-effective
- Solution:
 - blocking crossbar switching



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3.4 Crossbar Switch Configurations

□ Blocking crossbar switches

- Aiming to reduce the number of crosspoint switches.
- Category: Single stage / Multistage
- Methodology:
 - Two subscribers share one vertical bar
 - ✓ Number of bars reduced
 - ✓ Number of crosspoint switches remains the same.
 - All subscribers share a number of vertical bars
 - ✓ Numbers of bars and crosspoint switches reduced



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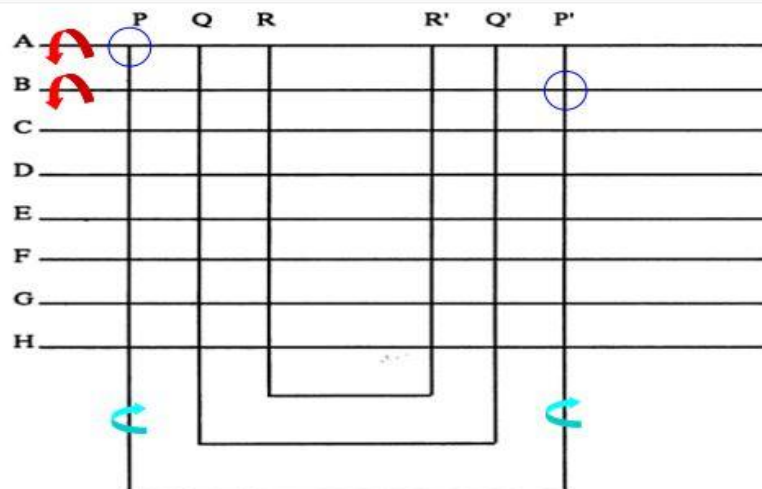


Fig. 3.9 Blocking crossbar switch.



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Blocking crossbar switches

□ Scheme 1

■ 2NK switches

- N denotes the number of subscribers
- K is the number of simultaneous connections
- Operate four bars to establish a connection (e.g. A-B).
 - Energize horizontal bar A
 - Energize free vertical bar P
 - De-energize horizontal bar A
 - Energize horizontal bar B
 - Energize free vertical bar P' (associated with P)
 - De-energize horizontal bar B



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Blocking crossbar switches

□ Scheme 2

■ NK switches

- N denotes the number of subscribers
- K is the number of simultaneous connections
- Operate three bars to establish a connection (e.g. A-B).
 - Energize horizontal bar A and B
 - Energize free vertical bar P
 - De-energize horizontal bar A and B

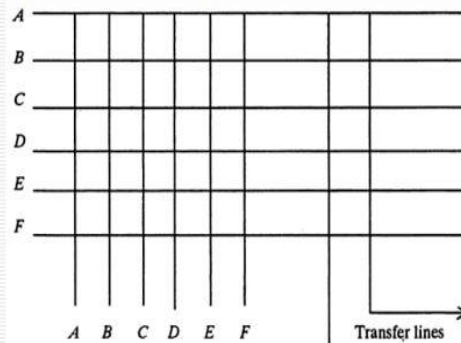


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Transfer line support

- Locally nonblocking and externally blocking



$$S = N(N+L)$$

(a) Locally nonblocking and externally blocking

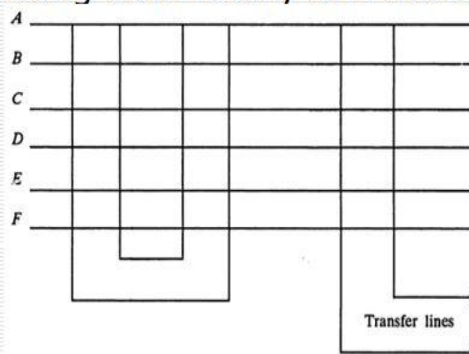


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Transfer line support

- Blocking both locally and externally



$$S = N(2K+L)$$

K (number of simultaneous call supported locally)

(b) Blocking both locally and externally

Fig. 3.10 Crossbar switches with transfer lines.



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3.5 Crosspoint Technology

- ❑ The cost of crossbar system increases in proportion to the number of crosspoints.
- ❑ Challenges
 - Reduction of the size of a crosspoint
 - Reduction of the cost of a crosspoint
 - Improvement of switching time



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3.5 Crosspoint Technology

❑ Category of crosspoint technology

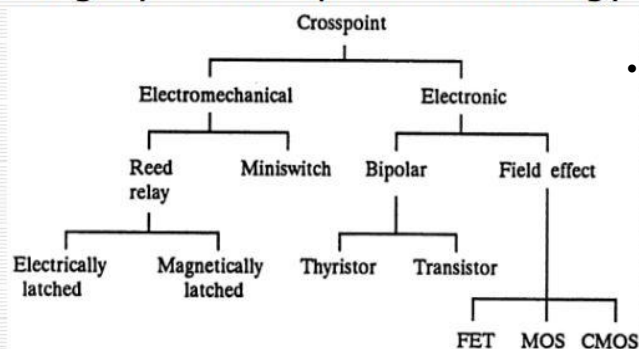


Fig. 3.12 Crosspoint technology.

- Reed Relay eliminate the mechanical motion of the Bars in a Crossbar system which increases the operation life time



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